

Assignment 5: Lenia – An artificial life

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Evaluation We use different versions to evaluate performance metrics.

Row distribution (1 proc) • $O(n^2)$ nested loop • Sequential execution • Single-process baseline Baseline: $\approx 0.08\times$ vs V2	Row distribution (16 procs) • $O(n^2)$ nested loop • Each pair counted twice • 16-process scaling Reference: $1\times$	Block distribution (16 procs) • $O(n^2)$, unique pairs only • Newton's 3rd law • Multi-core via OpenMP Reference: $1\times$
Block distribution + DT (16 procs) • Row & box decomposition • Localized particle blocks • Shared memory caching $\approx 4\times$ vs V2	GPU - NCCL (16 procs) • MPI derived data types • Non-contiguous packing • GPU-aware MPI transfer $\approx 4.5\times$ vs V2	

Table 1: Summary of all versions. Speedups at $N = 1000$ steps, relative to V2 (OpenMP CPU).

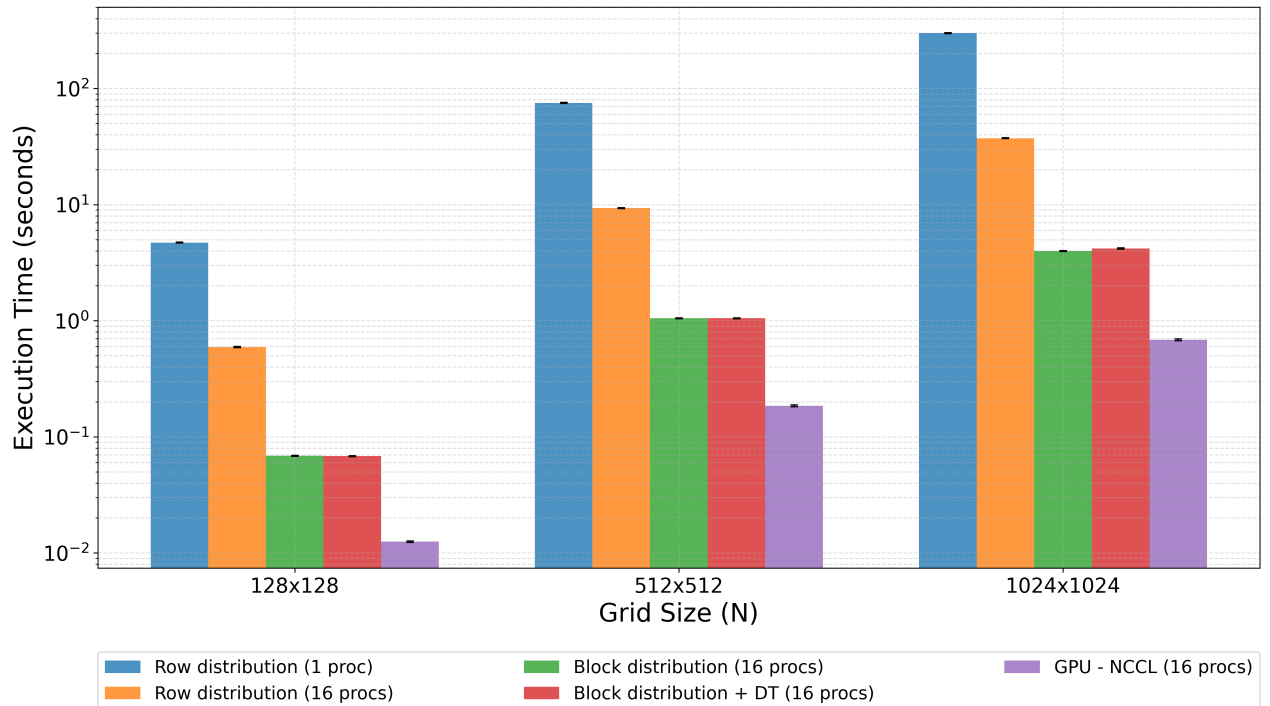


Figure 1: Performance comparison of the aforementioned optimization layers.

Now, we implement the row-wise distribution. Increasing the width of the border area exchanged between processes and performing additional computations on this border allows us to exchange data only every second, third, etc. iteration. The results are presented in the following image:

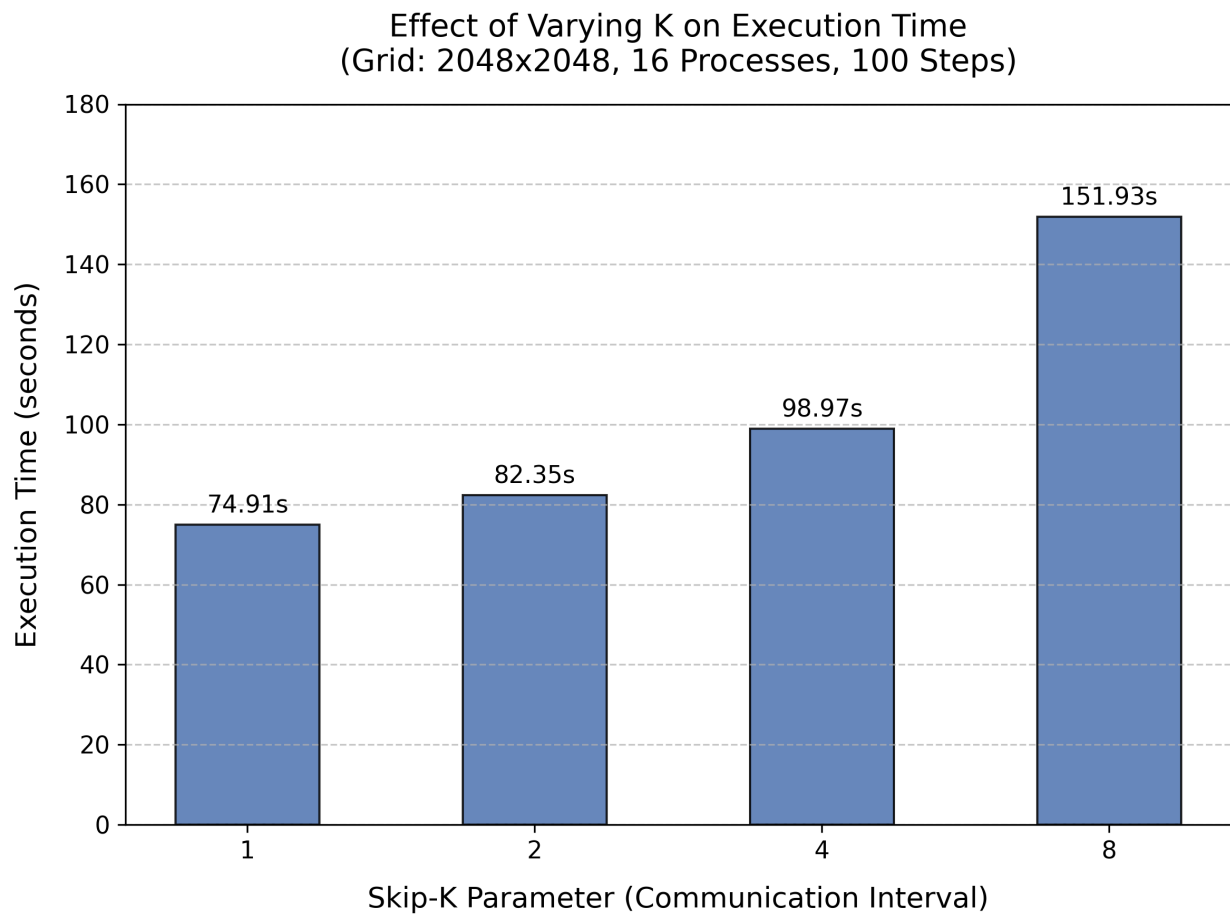


Figure 2: Performance comparison of the aforementioned optimization layers.